Deliverable 1: Entity-Relationship Diagram & Conceptual Design

Project: Intelligence Fusion Platform • **Course:** DBMS Mini Project (UE24CS395)
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1. Executive Conceptual Architecture & Polyglot Model

The **Intelligence Fusion Platform** model fuses two distinct database paradigms: a 22-table normalized relational core (PostgreSQL) and a flexible document store (MongoDB). Relational modeling ensures strict referential integrity, ACID compliance, and foreign-key cascades for evidence chain of custody and case dossiers. The document model manages polymorphic telemetric artifacts (CCTV logs, phone extractions, and encrypted chats).

2. Relational Entity Specifications

Entity	Type	Primary Key	Key Foreign Attributes	Cardinality & Role
cases	Strong	case_id (VARCHAR)	lead_officer_id (FK -> officers)	1:N with evidence, M:N with people/orgs
people	Strong	person_id (VARCHAR)	national_id (UNIQUE), primary_org_id	M:N with cases, vehicles, devices
organizations	Strong	organization_id	registration_number (UNIQUE)	M:N with cases, people
vehicles	Strong	vehicle_id (VARCHAR)	vin (UNIQUE), license_plate	M:N with cases, people
devices	Strong	device_id (VARCHAR)	imei (UNIQUE), mac_address	M:N with cases, people
locations	Strong	location_id (VARCHAR)	latitude, longitude, postal_code	M:N with cases, evidence scenes
evidence	Strong	evidence_id (VARCHAR)	case_id (FK), collected_by (FK)	1:N with custody audit events
evidence_custody	Weak/Log	custody_id (INT PK)	evidence_id (FK), officer_id (FK)	N:1 with evidence; audit trail
officers	Strong	officer_id (VARCHAR)	badge_number (UNIQUE)	1:N with cases, custody, activities
case_activity	Weak/Log	activity_id (INT PK)	case_id (FK), officer_id (FK)	N:1 with cases; chronological timeline

3. Associative Many-to-Many Bridge Tables

To satisfy **Boyce-Codd Normal Form (BCNF)** and eliminate repeating groups, 12 dedicated associative junction tables enforce referential constraints with composite primary keys and cascading rules:

Bridge Table	Primary Key (Composite)	Foreign Constraints	Investigative Semantics
case_people	(case_id, person_id)	ON DELETE CASCADE on both FKs	Role: Suspect, Witness, Informant, Victim
case_organizations	(case_id, organization_id)	ON DELETE CASCADE	Role: Front Company, Target, Shell Entity
case_vehicles	(case_id, vehicle_id)	ON DELETE CASCADE	Role: Getaway, Transport, Tracked Asset
case_devices	(case_id, device_id)	ON DELETE CASCADE	Role: Seized, Wiretapped, Infiltrated
case_locations	(case_id, location_id)	ON DELETE CASCADE	Role: Crime Scene, Safehouse, Cache
person_organizations	(person_id, organization_id)	ON DELETE CASCADE	Position: Director, Beneficiary, Courier

person_vehicles	(person_id, vehicle_id)	ON DELETE CASCADE	Association: Owner, Registered Driver, Renter
person_devices	(person_id, device_id)	ON DELETE CASCADE	Association: Primary User, Contact List
evidence_people	(evidence_id, person_id)	ON DELETE CASCADE	Association: Seized From, Attributed Person
evidence_vehicles	(evidence_id, vehicle_id)	ON DELETE CASCADE	Association: Recovered In / Transport Asset
evidence_devices	(evidence_id, device_id)	ON DELETE CASCADE	Association: Digital Media Extraction Source
evidence_locations	(evidence_id, location_id)	ON DELETE CASCADE	Association: Recovery / Chain of Custody Site

4. Editable Mermaid Source & Repository Artifacts

The editable diagram source is persisted in docs/er-diagram.md and docs/diagrams/er-diagram.mmd. It visually documents entity cardinalities (1:1, 1:N, M:N) and cross-database linkage between PostgreSQL entity UUIDs and MongoDB document collections.